

Exponential Growth and Decay Models in Differential Equations

Converted from Handwritten PDF

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$$P'(t) - rP(t) = 0, \text{ } r \text{ a real number}$$

is a 1st order, linear homogeneous DE.

Rewrite the DE as $P'(t) = rP(t)$.

$$\text{and } \frac{1}{P(t)}P'(t) = r, \text{ } P \neq 0$$

which is a separable DE.

Every 1st order linear homogeneous DE is also separable. (Not all separable DE is linear, for example, $y' = y^2(t)$.)

$$P' = rP$$

We know that the general solution is $P(t) = ke^{rt}$

and the integration constant k , is determined by an initial condition.

In this video we consider two applications that can be described by such an IVP:
 $y'(t) = r \cdot y(t), y(t_0) = y_0$

Exponential growth (if $r > 0$) or exponential decay (if $r < 0$) models.

1 Application

Let $P(t)$ denote a population size at a time t .

- Assume that the population only changes due to births and deaths (no immigration, or emmigration, for example) and
- that the rate of change of the population is proportional to the population size) $P' \propto P$.

Then $P'(t) = r P(t)$ where

r is the constant growth rate and is equal to the

birth rate — death rate.

If the birth rate $>$ death rate, then $r > 0$ and the population will grow exponentially.

If the death rate $>$ birth rate, $r < 0$ and the population will decay exponentially.

Applications include - yeast in a petri dish. - the populations of a region - atoms in a radioactive material

In what follows we will discuss the models where $r > 0$ and the model where $r < 0$ respectively.

Radioactive decay:

Atoms of a radioactive material such as Uranium 238 are unstable, and will spontaneously transition to other elements.

This process is called **radioactive decay**.

Experiments by Rutherford (1927) and others concluded that the rate at which radioactive atoms disintegrate is proportional to the amount of radioactive material.

That is $Q' \propto Q$

where Q is the number of radioactive atoms at time t .

In this example there are only deaths (of atoms as it transitions into a different material)

Since the “birth rate is zero”, the growth rate = death rate < 0

If $d > 0$ denotes the death rate

$$r = -d$$

$$Q'(t) = -dQ(t)$$

and we know $Q(t) = ke^{-dt}$.

Note: The same ideas, and models, are used for very different applications, human population growth and radioactive decay.

let P denote the number of individuals in a population at time t .

let the growth rate $r > 0$

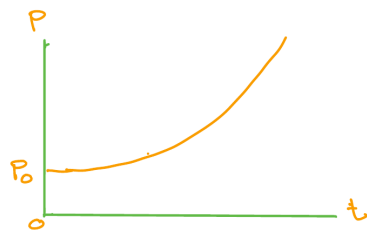
Then

$$P'(t) = rP(t), P(0) = P_0$$

The solution to this

IVP is

$$P(t) = P_0 e^{rt}$$



Assume $r = 0.02$,
and $P(0) = 3.6 \times 10^9$

Then we can calculate
 P at any time t .

For example

let Q denote the number of radio-active atoms in a material at time t .

let the decay rate $-d < 0$ ($d > 0$)

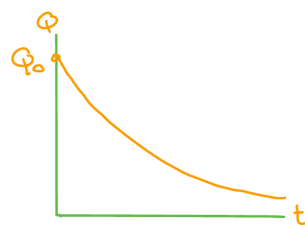
Then

$$Q'(t) = -dQ(t), Q(0) = Q_0$$

The solution to this

IVP is

$$Q(t) = Q_0 e^{-dt}$$



Assume $-d = -0.03$
and $Q(0) = 50$

Then we can calculate
 Q at any time t .

For example

Figure 1: Diagram or drawing from handwritten notes

1.1 Left Column

Let P denote the number of individuals in a population at time t .

Let the growth rate $r > 0$

Then

$$P'(t) = rP(t), \quad P(0) = P_0$$

The solution to this IVP is

$$P(t) = P_0 e^{rt}$$

{{FIGURE: Graph showing exponential growth curve, P vs t, starting at P_0 and increasing}}}

Assume $r = 0.02$, and $P(0) = 3.6 \times 10^9$

Then we can calculate P at any time t . For example

1.2 Right Column

Let Q denote the number of radioactive atoms in a material at time t .

Let the decay rate $-d < 0$ ($d > 0$)

Then

$$Q'(t) = -dQ(t), \quad Q(0) = Q_0$$

The solution to this IVP is

$$Q(t) = Q_0 e^{-dt}$$

{{FIGURE: Graph showing exponential decay curve, Q vs t, starting at Q_0 and decreasing}}}

Assume $-d = -0.03$ and $Q(0) = 50$

Then we can calculate Q at any time t . For example

1.3 Left Column

$$P(5) = P_0 e^{5r}$$

$$= 3.6 \times 10^9 e^{0.02 \times 5}$$

$$= 3.98 \times 10^9$$

(calculator.)

Note The estimated growth rate of the world population between 1960 and 1970 was $r = 0.02$ and the estimated population size in 1965 was 3.6 billion.

$P(5)$ then approximates the world population at $1965 + 5 = 1970$.

1.4 Right Column

$$Q(20) = 20.3.$$

Check that you get the same result.

If you know Q at two different times, you can calculate $-d$, for example, suppose you know $Q(t_1)$ and $Q(t_2)$, then

$$Q(t_1) = Q_0 e^{-dt_1}$$

and

$$Q(t_2) = Q_0 e^{-dt_2}$$

$$\frac{Q(t_1)}{Q(t_2)} = e^{-d(t_1 - t_2)}$$

from which you can find $-d$.

$$\ln \left(\frac{Q(t_1)}{Q(t_2)} \right) = -d(t_1 - t_2)$$

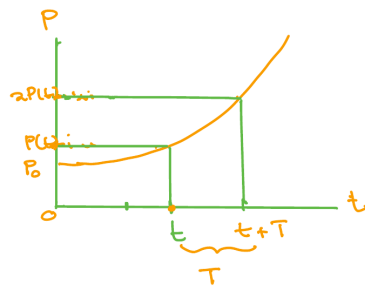
$$-d = \frac{\ln \left(\frac{Q(t_1)}{Q(t_2)} \right)}{(t_1 - t_2)} \rightarrow$$

Common questions are:

How long does it take the population to double?

The doubling time T satisfies

$$P(t+T) = 2P(t)$$



$$P(t+T) = 2P(t)$$

$$P_0 e^{r(t+T)} = 2P_0 e^{rt}$$

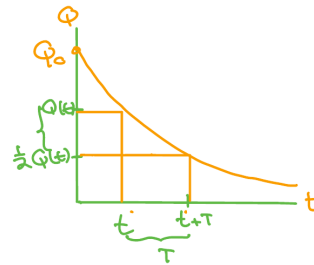
$$e^{rT} = 2$$

$$T = \frac{\ln 2}{r}$$

What is the half-life of a material?

The half-life T satisfies

$$Q(t+T) = \frac{1}{2}Q(t)$$



$$Q(t+T) = \frac{1}{2}Q(t)$$

$$Q_0 e^{-d(t+T)} = \frac{1}{2}Q_0 e^{-dt}$$

$$e^{-dT} = \frac{1}{2}$$

$$e^{dT} = 2$$

$$T = \frac{\ln 2}{d}$$

What is the longterm behavior

$$\lim_{t \rightarrow \infty} P(t) = P_0 e^{rt} = \infty$$

$$\lim_{t \rightarrow \infty} Q(t) = \lim_{t \rightarrow \infty} Q_0 e^{-dt} = 0$$

Figure 2: Diagram or drawing from handwritten notes

2 Common questions are:

How long does it take the population to double?	What is the half-life of a material?
The doubling time T satisfies $P(t+T) = 2P(t)$	The half-life T satisfies $Q(t+T) = \frac{1}{2}Q(t)$

{{FIGURE: Left graph showing exponential growth curve P vs t , with points marked at t and $t+T$ on x-axis, and P_0 , $P(t)$, $2P(t)$ on y-axis, with interval T indicated}}

{{FIGURE: Right graph showing exponential decay curve Q vs t , with points marked at t and $t+T$ on x-axis, and $\frac{1}{2}Q(t)$, $Q(t)$, Q_0 on y-axis, with interval T indicated}}

$$P(t+T) = 2P(t)$$

$$P_0 e^{r(t+T)} = 2P_0 e^{rt}$$

$$e^{rT} = 2$$

$$T = \frac{\ln 2}{r}$$

$$Q(t+T) = \frac{1}{2}Q(t)$$

$$Q_0 e^{-d(t+T)} = \frac{1}{2}Q_0 e^{-dt}$$

$$e^{-dT} = \frac{1}{2}$$

$$e^{dT} = 2$$

$$T = \frac{\ln 2}{d}$$

2.1 What is the longterm behavior

$$\lim_{t \rightarrow \infty} P(t) = P_0 e^{rt}$$

$$= \infty$$

$$\lim_{t \rightarrow \infty} Q(t)$$

$$= \lim_{t \rightarrow \infty} Q_0 e^{-dt} = 0$$

3 Summary:

- Different applications that can be modeled by $y' = ay$.
- Solutions for $y' = ay$, $y(t_0) = y_0$
using separation of variables or integrating factor, μ .
- Answer questions such as
 - $y(t) = ?$
 - Given $y(t_1)$, $y(t_2)$, calculate a or
 - Doubling time or half-life ($a > 0$) ($a < 0$)
- $\lim_{t \rightarrow \infty} y(t)$

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- Work through Examples 1, 2, 3 in Section 2.4.
 - You can now do Exercises 1, 2, 8 in Section 2.4.